

NETFLIX DATA ANALYSIS

PROJECT REPORT

Using Excel and Power BI

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Domain: Entertainment, Media & OTT Analytics

Table of Contents

1. Executive Summary
2. Problem Statement
3. Project Objectives and Business Use Cases
4. End-to-End Methodology
5. Phase 1 – Excel Data Cleaning and Preparation
6. Phase 2 – Power BI Data Modeling and Visualization
7. Dashboard Page 1 – Overview
8. Dashboard Page 2 – Content & Geographic Analysis
9. Dashboard Page 3 – Customer & Subscription Analysis
10. Dashboard Page 4 – Rating & Engagement Analysis
11. Key Business Insights
12. Business Recommendations
13. Conclusion
14. Technical Appendix – Dataset Profile

1. Executive Summary

The Netflix Data Analysis project is an end-to-end business intelligence solution designed to transform Netflix customer and content-related data into actionable insights. The project follows the supplied project specification, which emphasizes data cleaning, exploratory analysis, customer behavior analysis, content performance, subscription engagement and interactive Power BI dashboard creation.

2. Problem Statement

Netflix generates large volumes of content and customer engagement data. Without structured analysis, it is difficult to identify which customer groups are most engaged, which subscription types generate higher viewing activity, how ratings are distributed, and where customer engagement differs across countries. The project therefore focuses on cleaning the data and building interactive Power BI views that support content, customer and subscription decisions.

3. Project Objectives and Business Use Cases

Objective	Analytical approach	Business value
Content popularity	Compare content using watch time, ratings and category-level views	Identify high-performing content
Customer demographics	Group customers by age and analyse distribution	Understand target audience
Subscription engagement	Compare Basic, Standard and Premium watch time	Evaluate subscription engagement
Ratings analysis	Study 1–5 rating distribution and average rating	Understand satisfaction patterns
Geographic analysis	Compare customer and engagement patterns by country	Support localization strategy
Trending content	Review recently added content and ratings	Support release planning
Dashboard reporting	Use slicers, KPIs and interactive visuals	Enable faster decisions

The business use cases and project objectives are derived from the supplied Netflix project specification.

4. End-to-End Methodology

Phase	Main activities	Output
1. Excel	Validate structure, check missing values, standardize fields, create age groups and perform initial summaries.	Cleaned analytical workbook
2. Power BI Model	Load the prepared data, establish relationships/fields and define measures for KPIs.	Analysis-ready semantic model
3. Visualization	Build KPI cards, trend charts, category charts, maps, rating visuals and slicers.	Interactive dashboard
4. Insight Generation	Interpret customer, subscription, rating and geographic patterns.	Business recommendations

Pipeline: Raw data → Excel preparation → Power BI model → Interactive dashboard → Business insights.

5. Phase 1 – Excel Data Cleaning and Preparation

The first phase prepares the customer ratings data for reliable analysis. The workbook contains 20,000 records and 10 columns. All listed columns are populated in the uploaded worksheet, so no blank values are present in the available customer-rating table. The Age Group field is already derived and supports demographic segmentation.

- Validate unique customer_id records and check show_id relationships.
- Check data types for age, watch_time_minutes and rating.
- Verify rating values follow the 1–5 scale.
- Standardize categorical values for gender, country and subscription type.
- Use the Age Group classification: 13–18, 19–30, 31–45, 46–60 and 60+.
- Prepare the cleaned workbook for Power BI import.

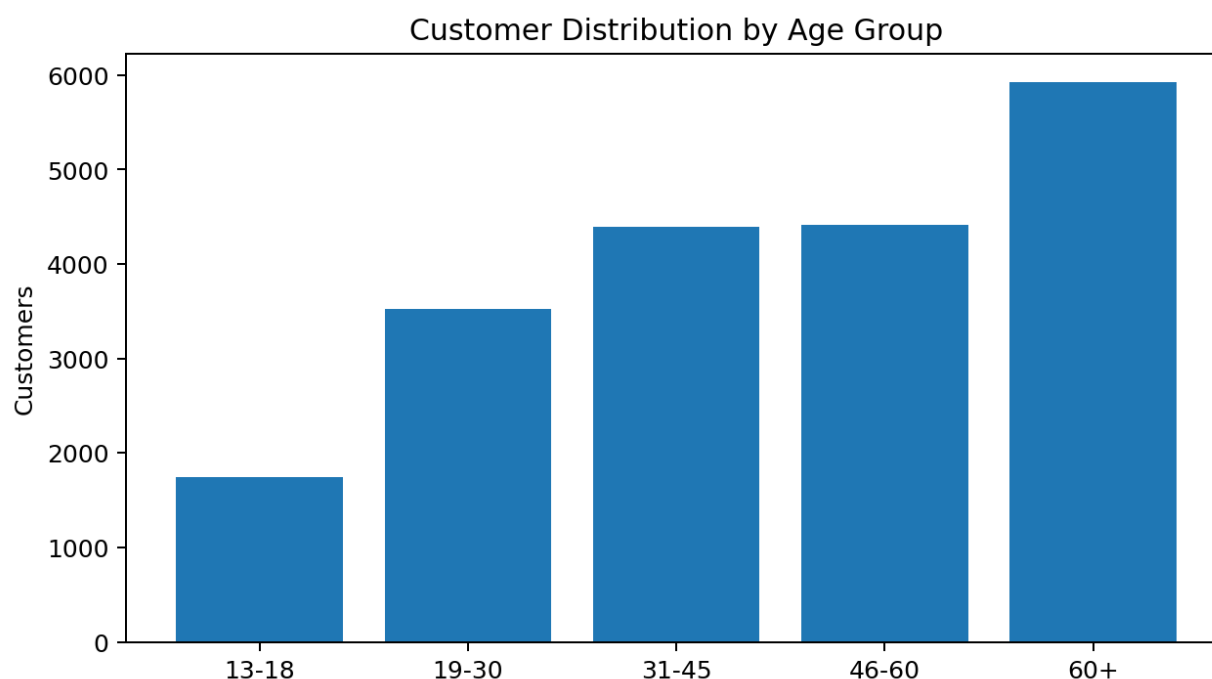


Figure 1. Customer distribution by age group.

The age-group distribution shows that the 60+ segment is the largest customer group, followed by 46–60 and 31–45.

6. Phase 2 – Power BI Data Modeling and Visualization

The uploaded Power BI project contains a Home page and four analytical pages. The report uses KPI cards, line and column charts, donut and bar charts, a filled map, scatter analysis, stacked charts, slicers and navigation buttons. These elements align with the project guideline to use relevant visualizations, filters and interactive reporting.

- Create KPI measures for customer count, watch time and average rating.
- Use slicers for country, age group and subscription type where applicable.
- Compare subscription plans using watch-time measures.
- Analyse geographic distribution using country-level visuals.
- Use rating and engagement visuals to identify customer behavior patterns.
- Maintain consistent navigation across the report pages.

7. Dashboard Page 1 – Overview

The Overview page acts as the executive landing page. The Power BI file contains four KPI card visuals followed by trend and category-oriented visuals. This page is intended to provide a fast summary of customer engagement and the major business indicators before users move to deeper analysis.

Customer Records	Total Watch Time	Average Watch Time	Average Rating
20,000	50,112,681 min	2,505.6 min	3.01 / 5

The Overview page is designed for quick management-level interpretation and interactive filtering.

8. Dashboard Page 2 – Content & Geographic Analysis

The second analytical page focuses on content and geographic comparison. The Power BI layout contains a bar chart, a filled map, a clustered bar chart and a donut chart. These visuals support country-level engagement analysis, content-category comparison and distribution analysis.

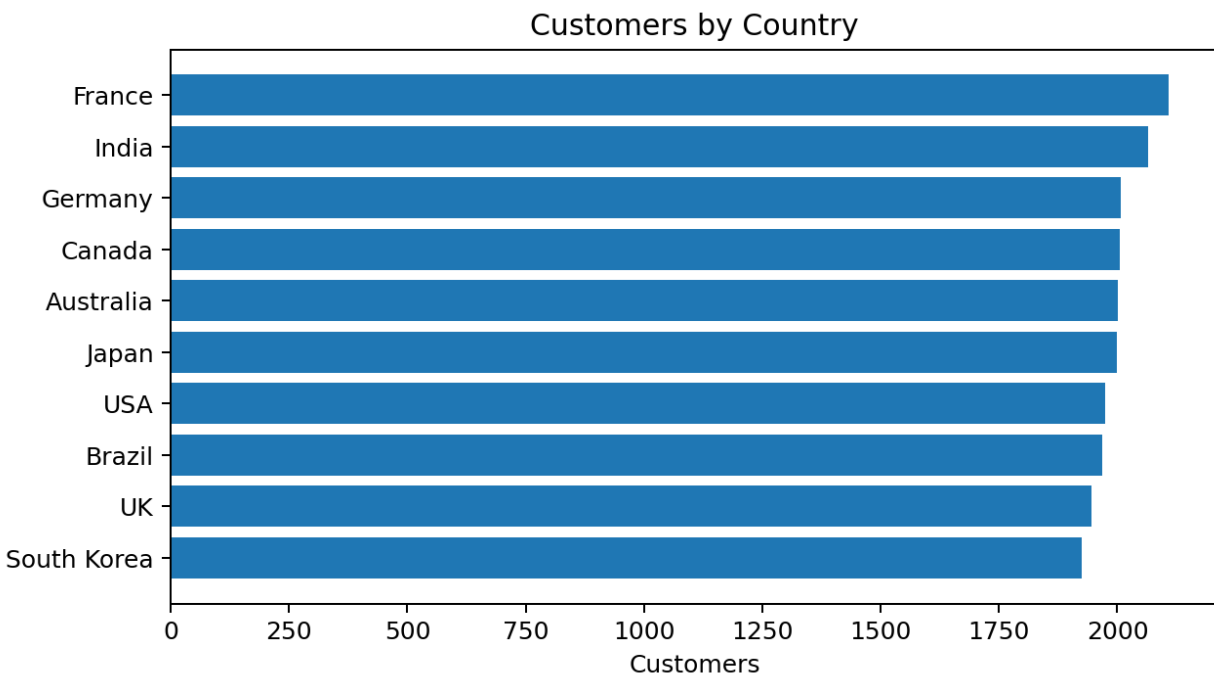


Figure 2. Customer distribution across the available countries in the uploaded customer dataset.

Among the available customer records, France has the highest customer count at 2,109, followed by India at 2,064 and Germany at 2,007. This supports country-level comparison as part of the localization strategy.

9. Dashboard Page 3 – Customer & Subscription Analysis

The third page focuses on customer and subscription behavior. The Power BI file contains clustered column and bar charts that can be used to compare demographic groups, subscription plans and customer engagement. The analysis below is calculated from the uploaded customer ratings workbook.

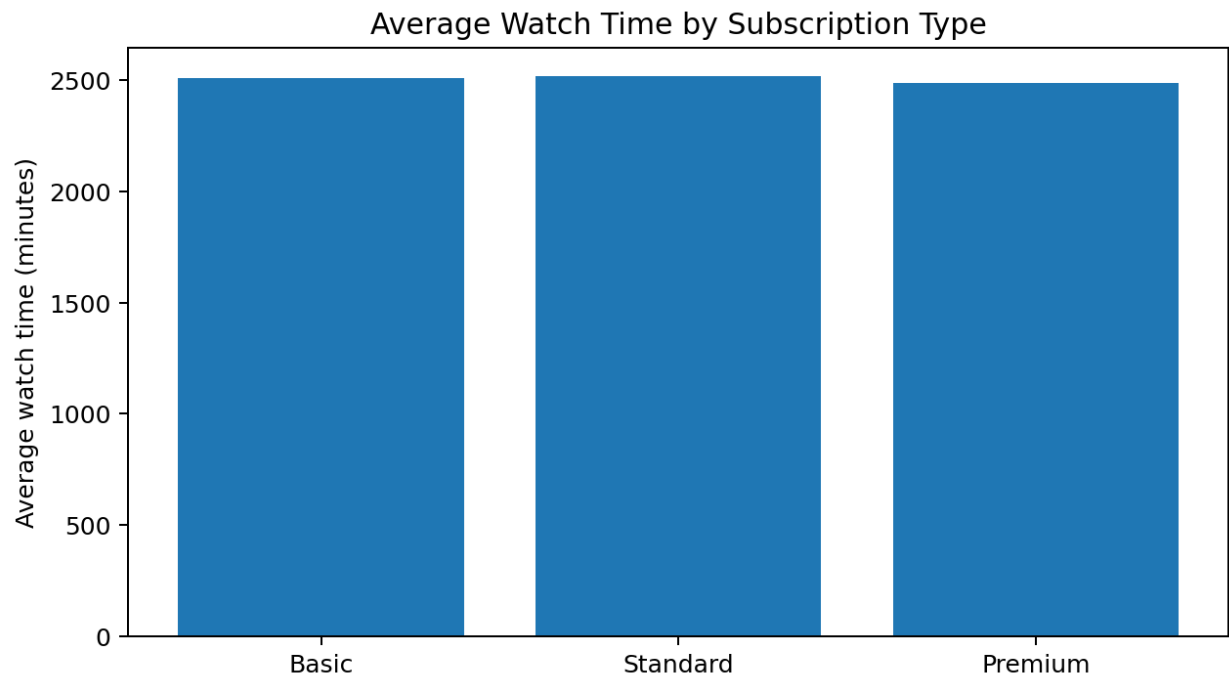


Figure 3. Average watch time by subscription type.

Subscription	Customers	Total watch time	Average watch time
Basic	6,595	16,559,037	2,510.8
Standard	6,856	17,268,939	2,518.8
Premium	6,549	16,284,705	2,486.6

Standard subscribers form the largest customer group (6,856) and also have the highest total watch time (17,268,939 minutes) and highest average watch time (2,518.8 minutes).

10. Dashboard Page 4 – Rating & Engagement Analysis

The fourth Power BI page contains a scatter chart and 100% stacked column/bar visuals. These are suitable for exploring relationships among ratings, watch time and categorical customer dimensions.

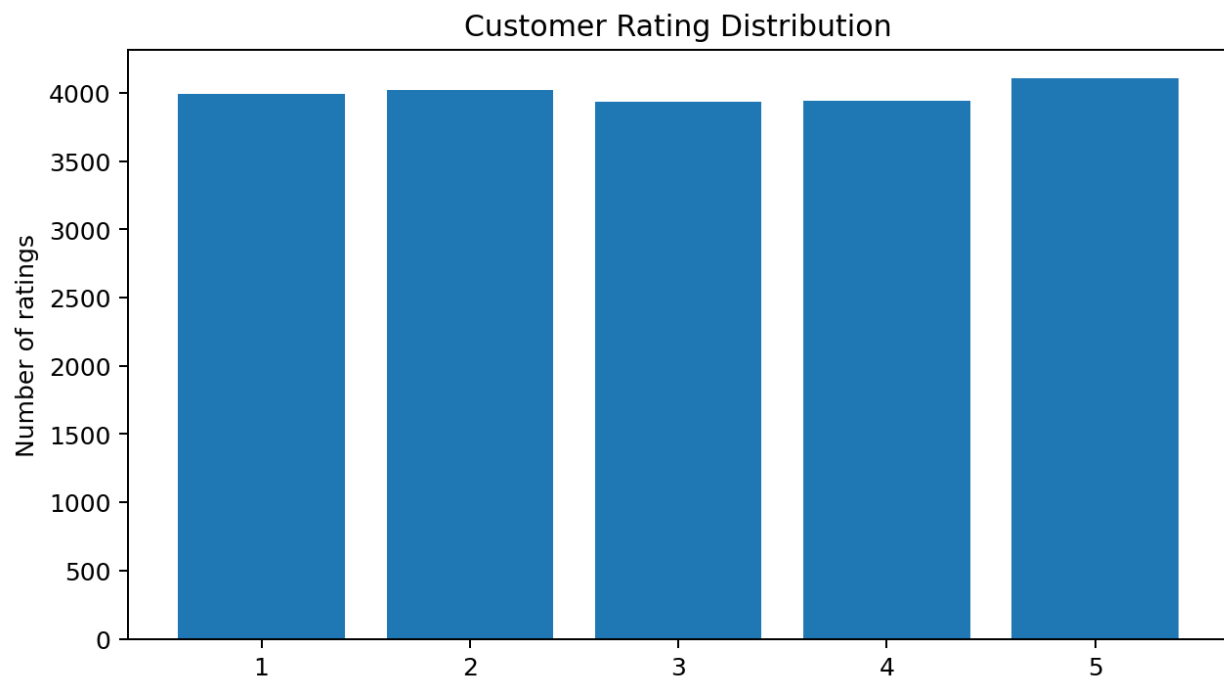


Figure 4. Distribution of customer ratings from 1 to 5.

The rating distribution is relatively balanced. Rating 5 is the most frequent with 4,108 records, while rating 3 is the least frequent with 3,935 records. The overall average rating is 3.01/5.

11. Key Business Insights

- The 60+ age group is the largest segment with 5,926 customers (29.63% of the dataset).
- Standard is the most represented subscription plan with 6,856 customers (34.28%).
- Standard subscribers have the highest total watch time at 17,268,939 minutes.
- The average watch time across all customer records is 2,505.6 minutes.
- The overall customer rating average is 3.01/5.
- Rating 5 is the most common individual rating with 4,108 records.
- France has the highest customer count among the ten available countries with 2,109 records.
- Age 60+ contributes the largest total watch time (14,712,614 minutes), although its average watch time per customer is lower than the younger groups.

12. Business Recommendations

- Prioritize engagement strategies for Standard subscribers because this plan shows the highest total and average watch time in the available customer dataset.
- Use age-specific personalization, especially for the 60+ group, which represents the largest customer population.
- Monitor the 13–18, 19–30 and 31–45 groups for engagement campaigns because their average watch times are higher than the 60+ group.
- Use country-level dashboard filters to identify markets with lower customer counts and compare them with ratings and watch-time patterns.
- Use the rating distribution together with watch time rather than ratings alone when evaluating content performance.

- Use the recent-content and release-month analyses specified in the project brief to support content release planning once the content-level dataset is connected.

13. Conclusion

The Netflix Data Analysis project successfully converts the supplied customer ratings data into an interactive business intelligence solution using Excel and Power BI. The analysis covers customer demographics, subscription engagement, watch time, ratings and geographic distribution, while the Power BI report adds interactive navigation, slicers and multiple analytical visualizations.

The results show that the 60+ age group is the largest customer segment, Standard is the strongest subscription group by watch-time engagement, and the overall rating level is close to 3 out of 5. The dashboard structure provides a practical foundation for deeper content-performance and localization decisions. Future analysis can be strengthened by combining the customer ratings table with the full Netflix Shows Dataset for title, genre, director, cast, release year and date-added analysis.

14. Technical Appendix – Dataset Profile

Metric	Value
Rows	20,000
Columns	10
Unique customers	20,000
Unique shows referenced	12,602
Countries	10
Subscription types	3
Minimum age	13
Maximum age	80s
Average age	46.63
Average watch time	2505.63 minutes
Total watch time	50,112,681 minutes
Average rating	3.01/5